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Executive Summary: 2024 CSU Systemwide Business Analytics Competition

Introduction

This project developed a predictive model to classify loan applications into lower-risk (approved) or higher-risk (rejected) categories. The primary goal was to maximize net profit while ensuring the accuracy and reliability of the predictions, utilizing comprehensive data cleaning and model development techniques**.**

Data Cleaning and Standardization

The dataset comprised 899,164 observations and 27 columns, consisting of 18 categorical and 9 numerical variables. The outcome variable was reclassified into two categories: lower risk (0) and higher risk (1). Data Cleaning involved extracting relevant values, grouping data, and removing missing or inaccurate observations. As detailed in the file labeled *1 - Data Cleaning,* irrelevant identifier columns, redundant columns, and those with unavailable data at the time of loan approval were excluded. Multicollinearity was addressed by removing additional columns. Visualizations were employed to assess both categorical and numerical variables, helping to understand the distribution and identify potential issues within the dataset. Classification trees were later used to assess feature importance, guiding the removal of non-predictive columns to enhance model efficiency.

Model Development

Dummy variables were created for all categorical variables. The dataset was split into a 60/40 training-validation ratio. Predictors were normalized as necessary, and hyperparameters were optimized using cross-validation. Weights were applied to the data to account for the high cost of misclassifying higher-risk loans.

The following models were developed using Python, with the Elastic Net variant of logistic regression implemented in R. For each model, the respective file labels are as follows:

- ***Classification Tree*** (File labeled *2 - Classification Trees*)
- ***k-Nearest Neighbors*** (***kNN***) (File labeled *3 – kNN Group Project*)
- ***Logistic Regression*** (***Logit Models***) (Files labeled *4 - Logistic Regression*)

- ***Discriminant Analysis*** (File labeled *5 - Discriminant Analysis*)
- ***Neural Networks*** (File labeled *6 - Neural Networks*)

Model performance was evaluated using accuracy scores, confusion matrices, and metrics such as gains and lift charts. Net profits were calculated to determine the optimal cutoff thresholds.

Model Results and Comparison

All models were compared, as detailed in the file labeled *7 - Model Comparison*, where ***Random Forest*** emerged as the most effective model, providing the highest net profit. Further analysis of the ***Random Forest*** model, including its performance metrics, propensities for validation set observations, and the optimal stopping point to achieve maximum net profit, is documented in the Python file labeled "*8 - Question 3"*.

Conclusion

This project successfully identified a predictive approach for loan approval through rigorous data cleaning and model development. The ***Random Forest*** model demonstrated superior performance, providing actionable insights for loan decision-making. Results and further implementation details are available in the accompanying Python files, with Elastic Net analysis documented in the R Markdown file**.**